

Recent progress of the long-wave infrared instrument for atmosphere gray extinction characterization within the StarDICE experiment

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MONTPELLIER



■ Content

1. **Why using an infrared thermal camera ?**
2. **Goals**
3. **Radiometric calibration**
4. **Overview of first observations**

Definitions and notations

- **LWIR** : Long-Wave Infrared band $\sim 7 - 14 \mu\text{m}$
- **Radiance** : radiant flux emitted, reflected, transmitted, or received by a surface, per unit solid angle per unit projected area = **L** in $\text{W}/\text{m}^2/\text{sr}$
- **Spatial noise** : standard deviation of a single complete image
- **Temporal noise** : standard deviation of a single pixel value on consecutive images for a supposed constant scene radiance
- **Cloud optical depth τ** : measure of the attenuation of light due to cloud particles along a given path
- **Radiometric** $\Rightarrow$ LWIR thermal camera (sky measurements)
- **Photometric** $\Rightarrow$ UV-VIS-NIR w/ CCD camera (stars measurements)

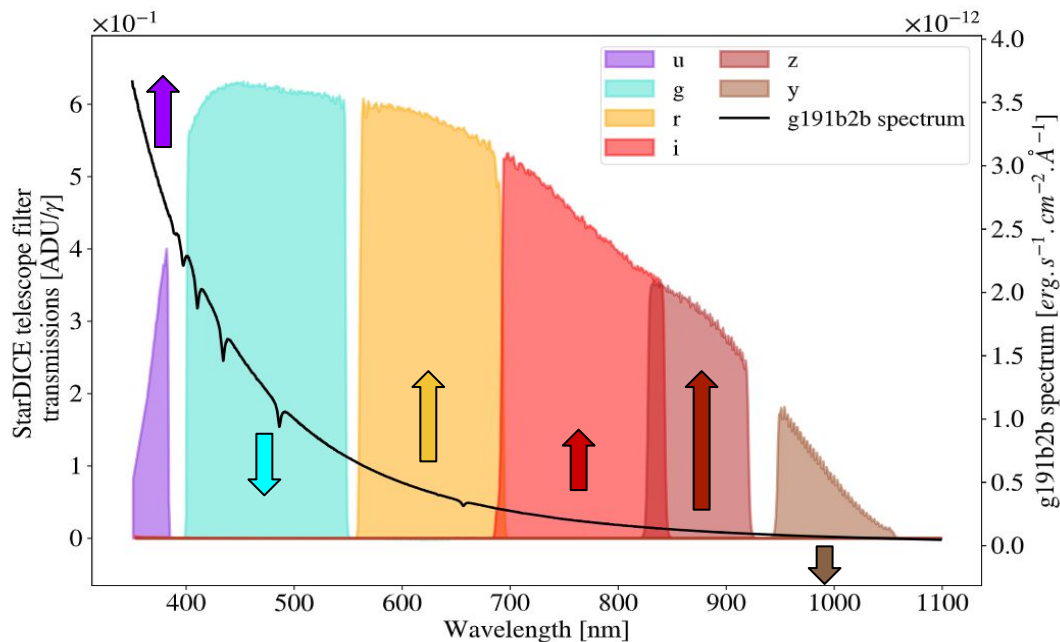
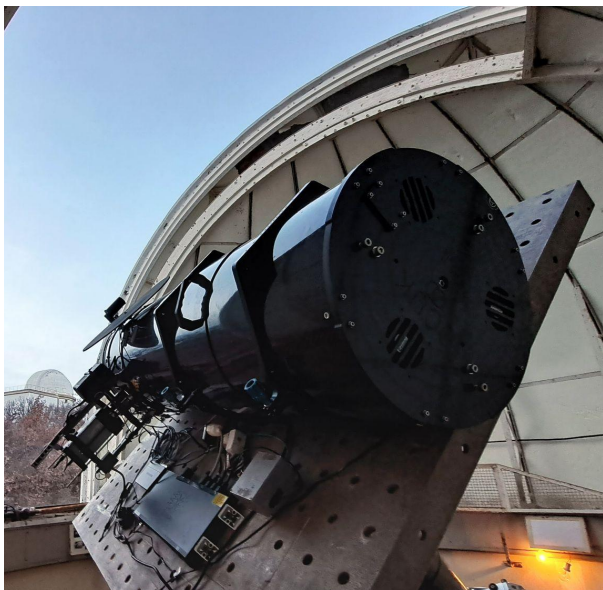
StarDICE metrology experiment

StarDICE is observing CALSPEC spectrophotometric standards

- Prior spectra given by CALSPEC
- Prior knowledge of filter transmissions (CBP + DICE)
- Prior atmosphere simulation

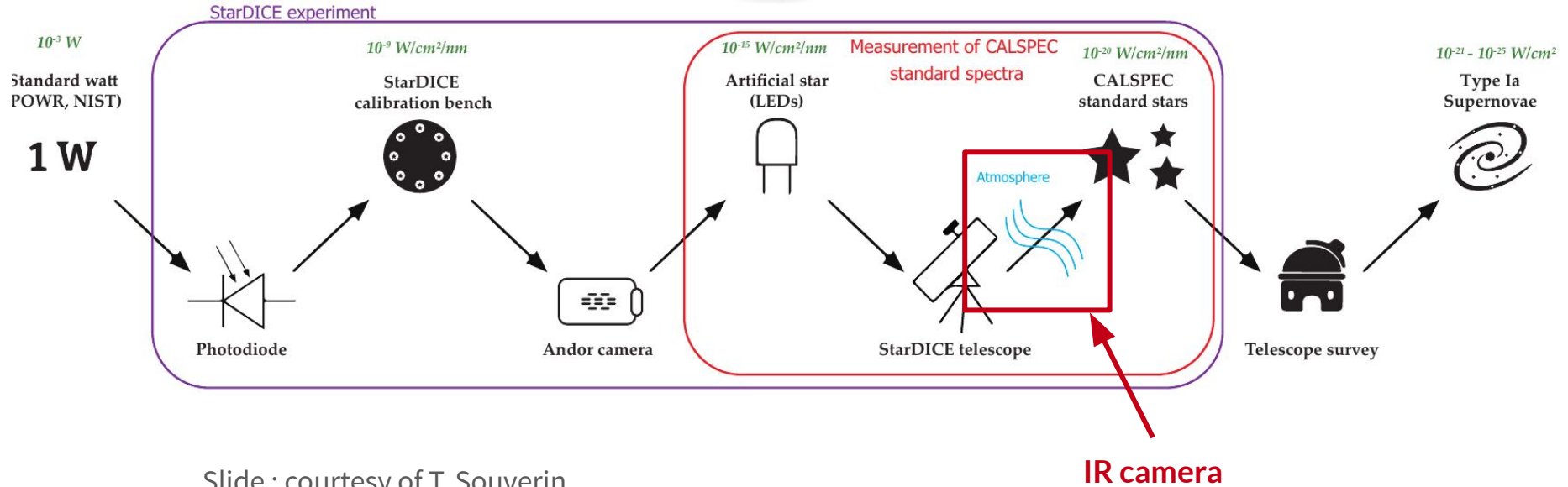
Slide : courtesy of T. Souverin

⇒ **Theory/Measurements to adjust the spectrum for each filter**



1. Why using an infrared thermal camera ?

1.1 StarDICE calibration transfer schematic



1.2 “Gray” extinction as a limiting systematic

$$F_X = A_{tel} \times t_{exp} \times \int_{\lambda} S_*(\lambda) \times T_X(\lambda) \times T_{atm}(\lambda) \times \frac{\lambda d\lambda}{hc}$$

→ CBP instrument, T. Souverin
presentation (8th November 2023)

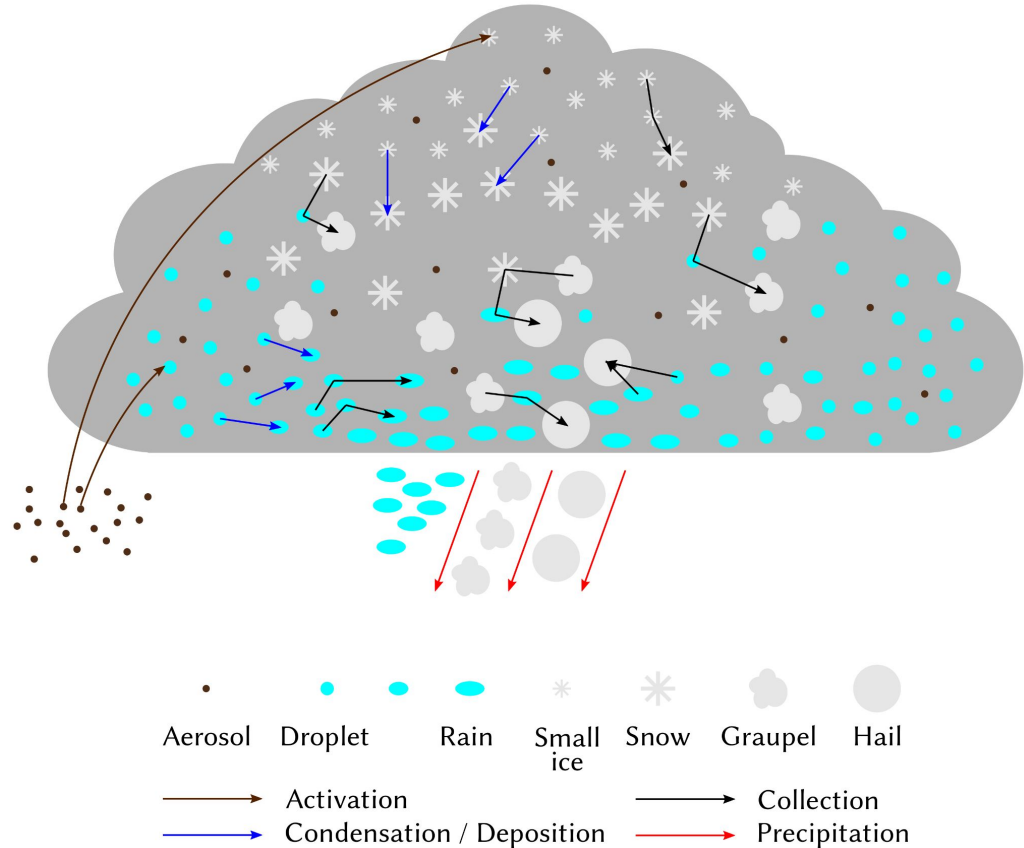
$$T_{atm}(\Theta) = T_{gray}(t, alt, az) \times \frac{T_{mol}(P_{atm}, t, alt, az, \lambda, O_3)}{\times T_{PWV}(PWV, t, alt, az, \lambda) \times \exp(-X \times \tau_{aer}(t, \lambda))}$$

- Empirical wavelength-independent “gray” normalization T_{gray} is fitted for each observation (Burke et al. 2010) → **insufficient as calibration is limited by a rapidly changing conditions with moving and evolving cloud cover + Simonyi telescope will observe during non-photometric conditions**
- Given by spectra + local measurements + radiative transfer simulations

1.3 What is causing “gray” extinction ?

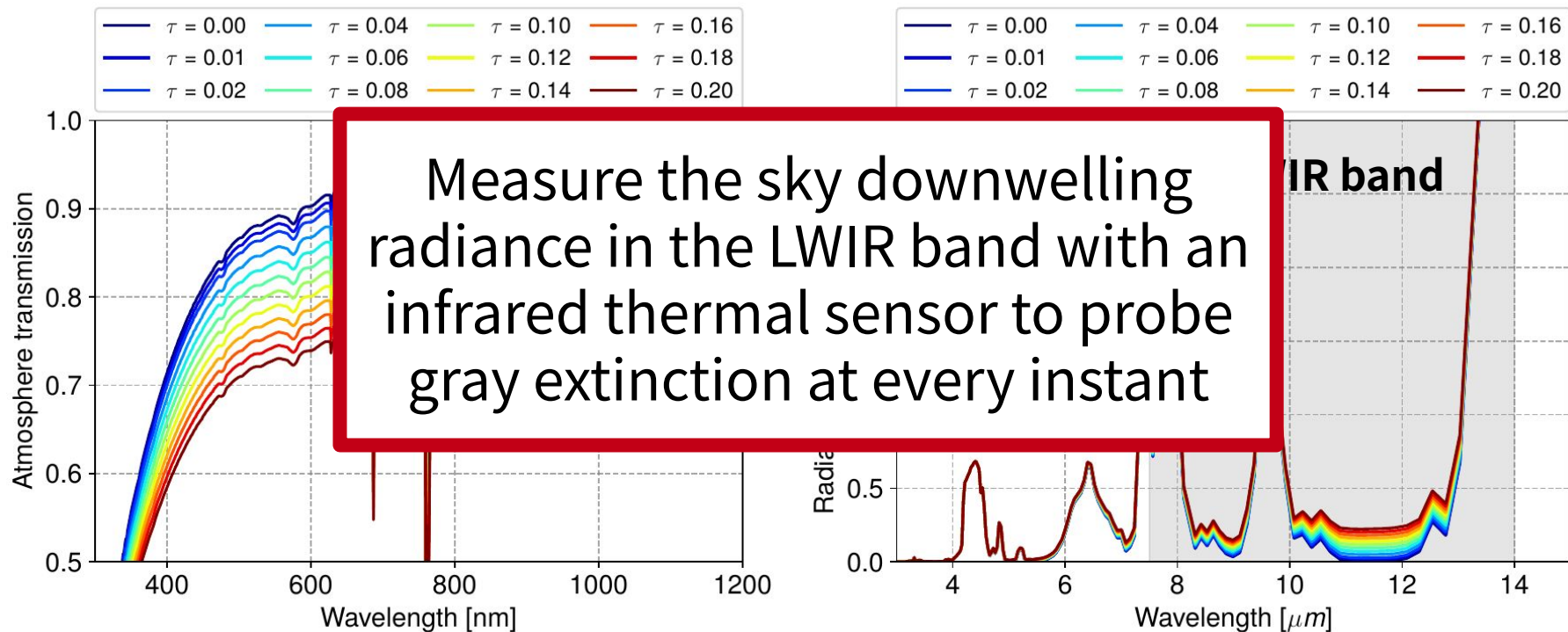
Ice crystals and water droplets in clouds :

- Produce shadows that appear as “gray” in UV-VIS-NIR (independent of wavelength) $\Rightarrow$ **decrease in atmosphere transmission** $\Rightarrow$ decrease in stellar source flux
- Produce thermal emission $\Rightarrow$ **increase in radiance / brightness temperature**



1.4 Effect on UV-VIS-NIR photometry and LWIR radiance

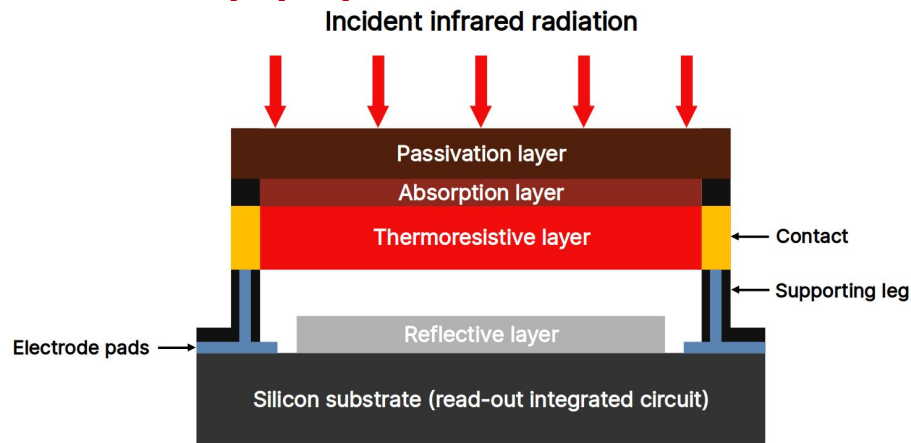
→ Measuring the sky radiance in LWIR band appears as a potential solution



Credits: Sylvie Dagoret-Campagne

1.5 Description of the instrument (1/2)

- 640 x 512 pixels uncooled microbolometers focal plane array
- Large FOV = 10.4 x 8.3 deg
- 8.33 Hz imaging rate $\Rightarrow$ ~ 80 GB/night
- Radiometrically calibrated radiance images in radiances $\text{W/m}^2/\text{sr}$

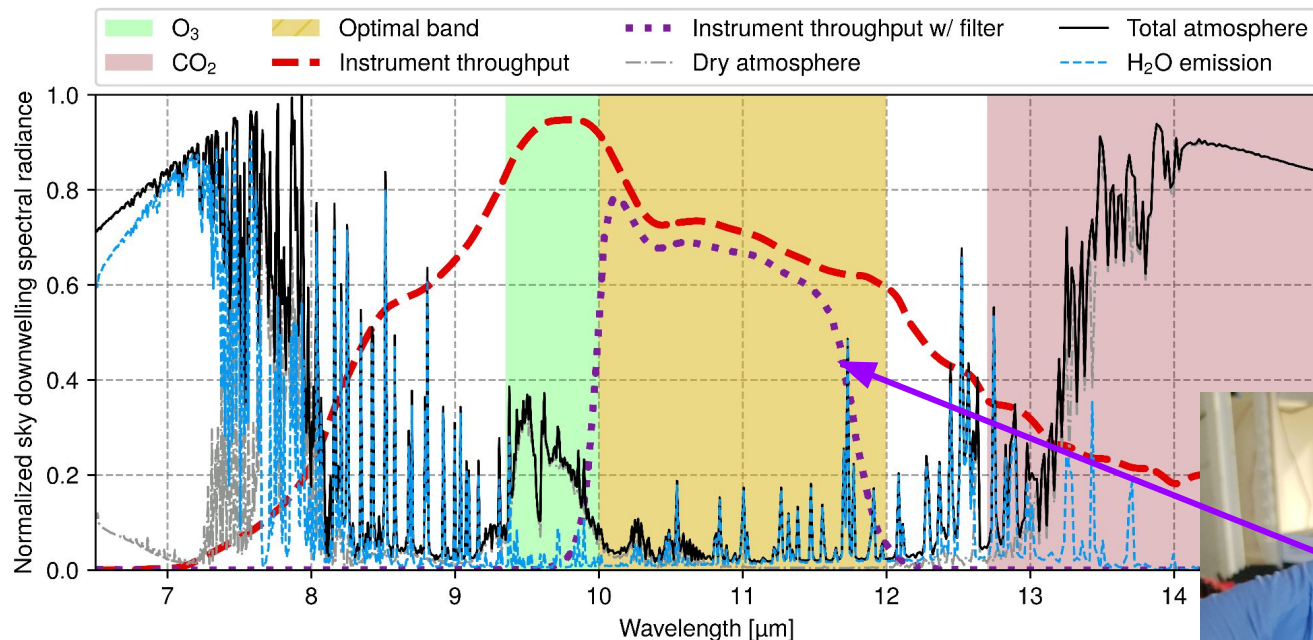


$$\Delta V = A \times L_{s2} + B(L_{pix})$$

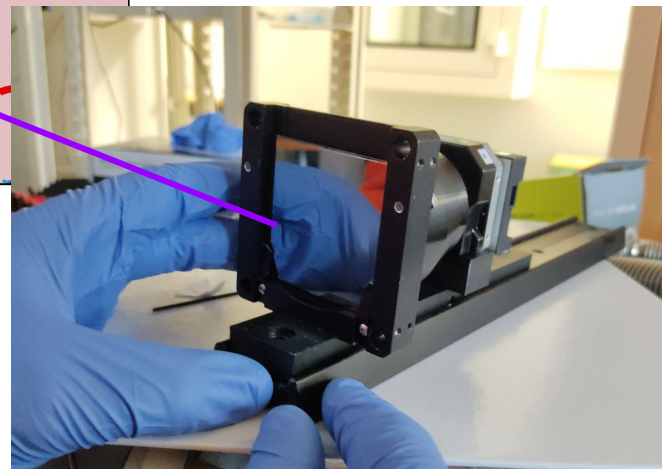
Labels above the equation: Voltage change (ΔV), Gain (A), Scene radiance (L_{s2}), Offset ($B(L_{pix})$).

Does definitely not work like a photon counting sensor !

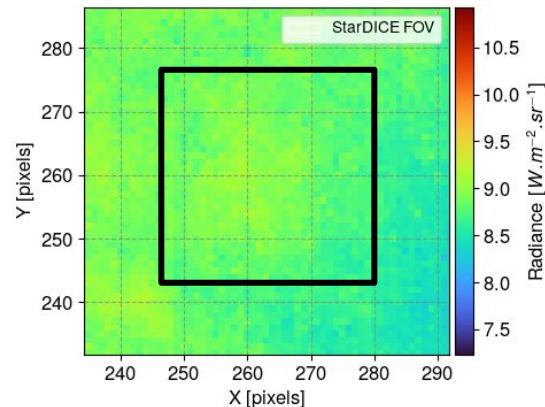
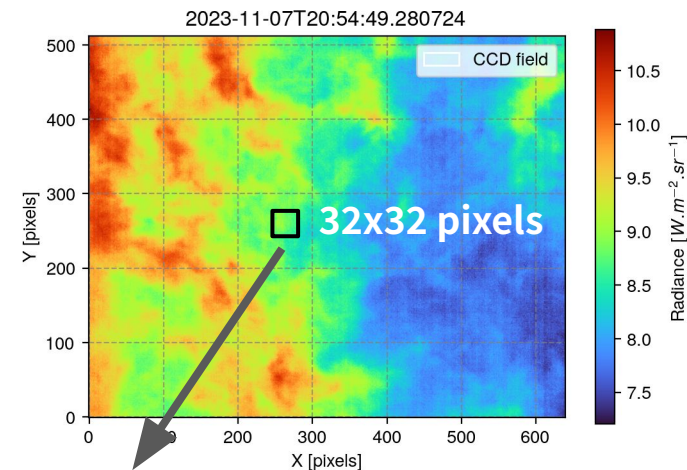
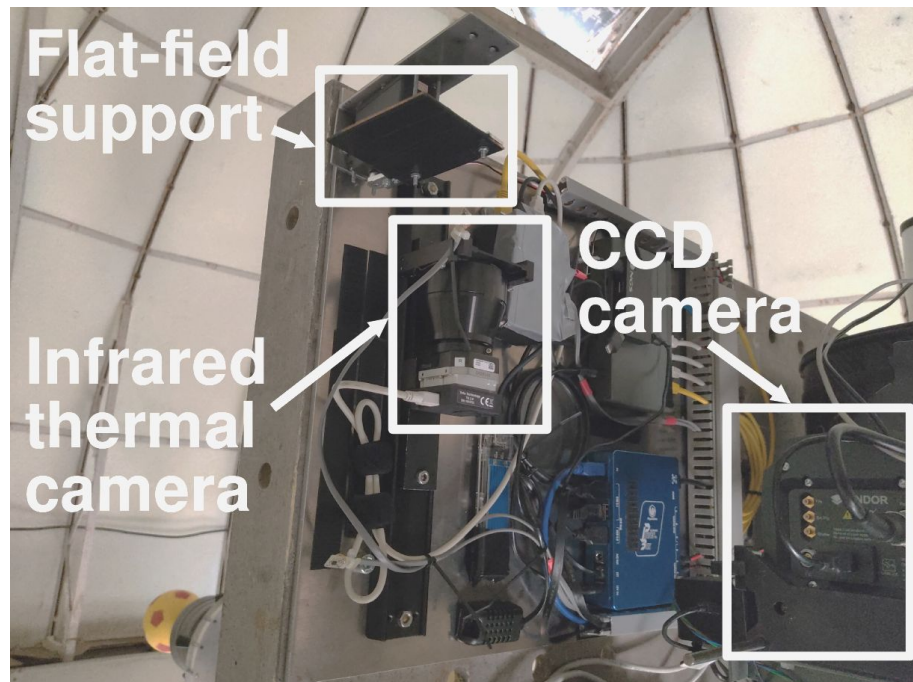
1.6 Description of the instrument (2/2)



Improve contrast by observing the **golden** area
⇒ dedicated 0-12 μm bandpass filter to isolate the optimal band
⇒ or excellent modeling of CO₂, O₃



1.7 Infrared instrument installed at Observatoire de Haute-Provence



2. Goals

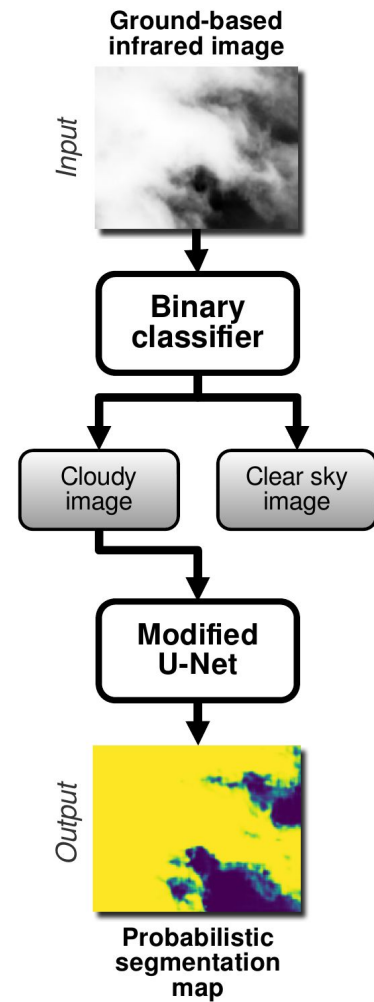
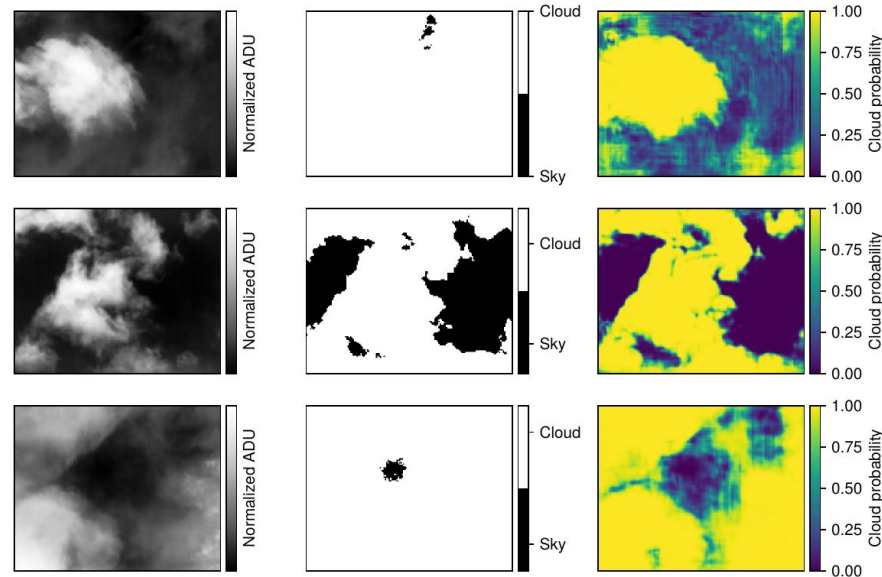
2.1 Short-term goal(s)

- Identify spectro/-photometric exposures affected by gray extinction / clouds
 - classification, cloud structure detection / image segmentation
 - flag them and remove them from the analysis

Infrared Radiometric Image Classification and Segmentation of Cloud Structure Using Deep-learning Framework for Ground-based Infrared Thermal Camera Observations

K. Sommer, W. Kabalan, R. Brunet
<https://doi.org/10.5194/egusphere-2024-101>

Atmosphere Measurement Techniques, European Geosciences Union (AMT Copernicus)



2.2 Long-term goal(s)

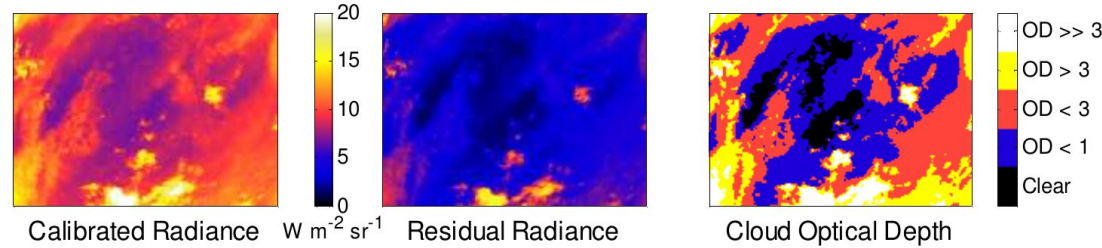
- Correct photometry for spurious variations

→ find a method to produce the quantitative metric T_{gray}

→ comparisons with simulations: sky background radiance removal to derive the gray extinction component

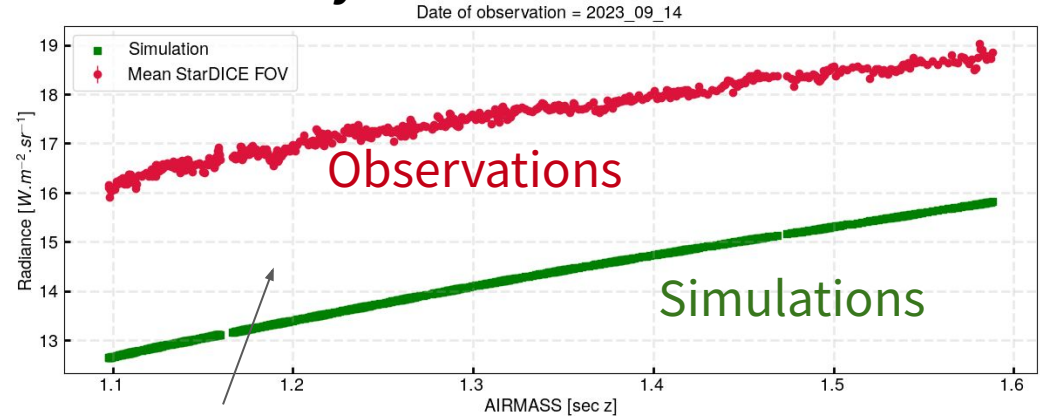
→ joint-analysis with other data: observed spectrum, lidar, AERONET to fit the atmosphere

→ correct sub-regions of CCD images



Source: Nugent et al. 2009

Sky thermal IR radiance



Discrepancy: calibration ? simulation ? both ?
→ work in progress

3. Radiometric calibration

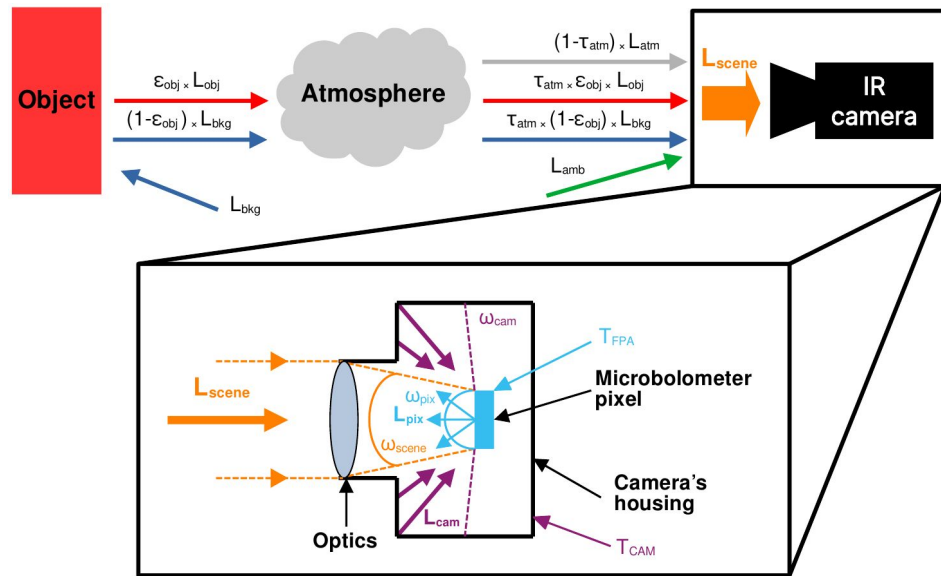
3.1 Motivation

- **Why calibrating the camera ?**

- Correct for camera gain and offset drifts
- Compare nights between each other
- Compare to simulations of sky radiance to evaluate the background level (e.g. libRadTran radiative transfer code)

- **How ?**

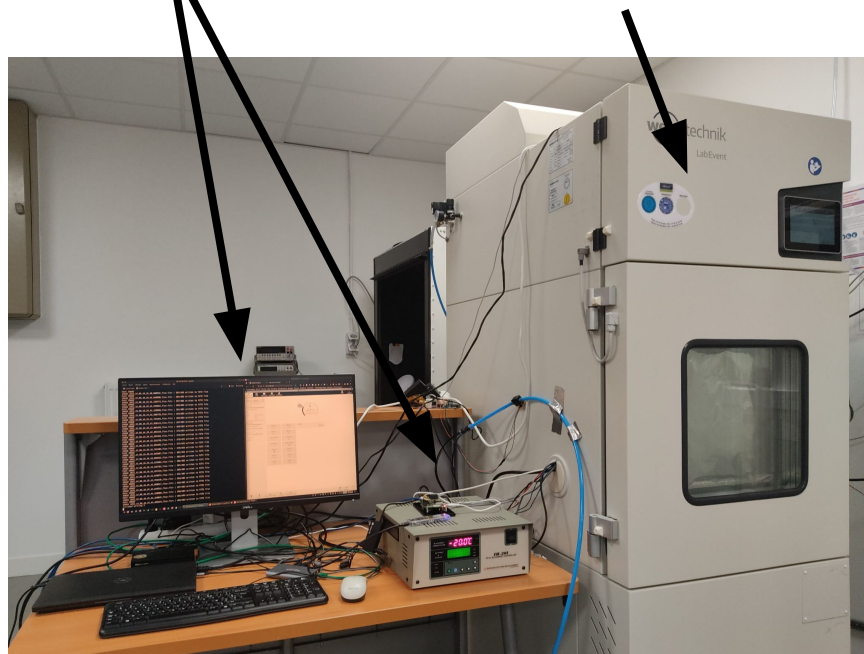
- Model the heat transfers occurring inside the camera + the observed scene
- Acquire data with a controlled thermal source (e.g. blackbody surface with known temperature and emissivity)
- Put the camera at different ambient temperatures inside a controlled area



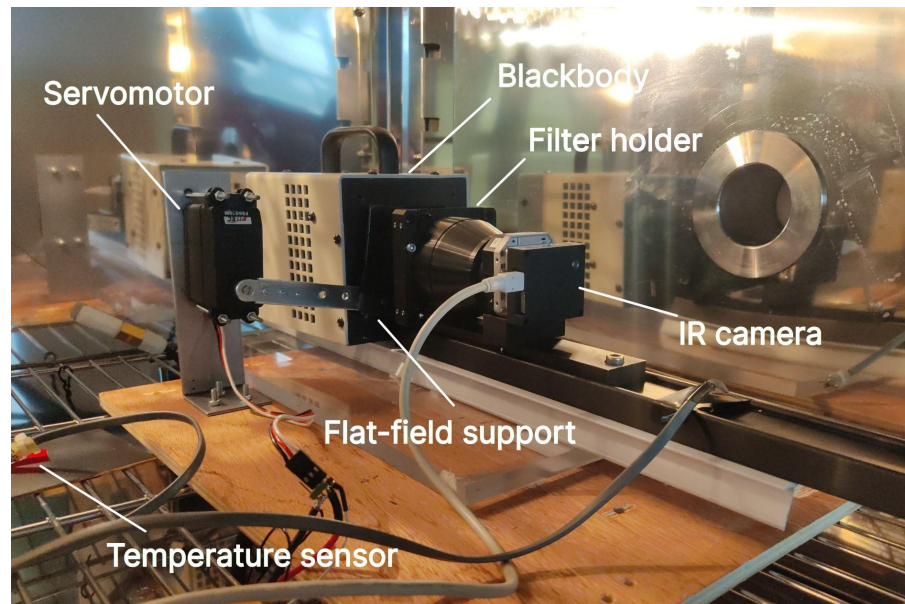
3.2 Experiments

With Sylvie Dagoret-Campagne and Marc Moniez
in December 2022

Control-command setup
Climatic chamber
for CERN ATLAS
experiment



+300k images for multiple camera and
low blackbody temperature setpoints

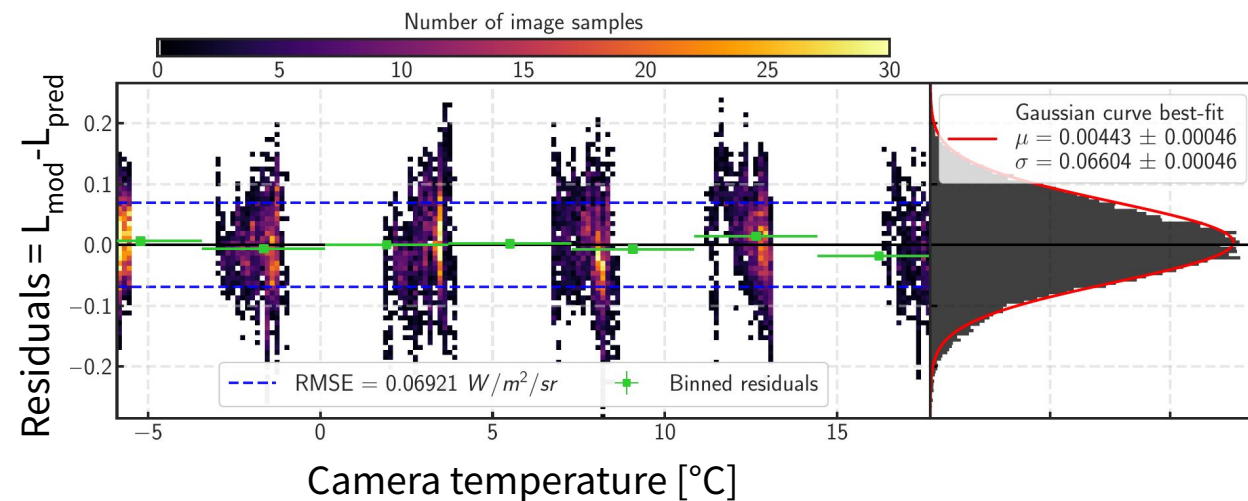


3.3 Results (preliminary)

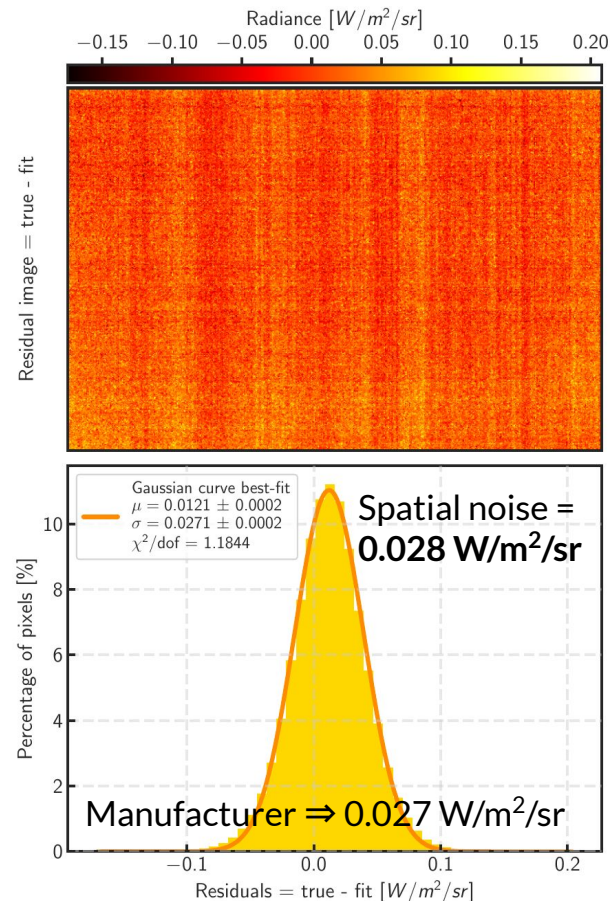
Pixel temporal noise $\Rightarrow \pm 0.10 \text{ W/m}^2/\text{sr}$

* Best calibration found in the literature : $\pm 0.36 \text{ W/m}^2/\text{sr}$
(Nugent, 2008, PhD)

K. Sommer *et al.* in prep

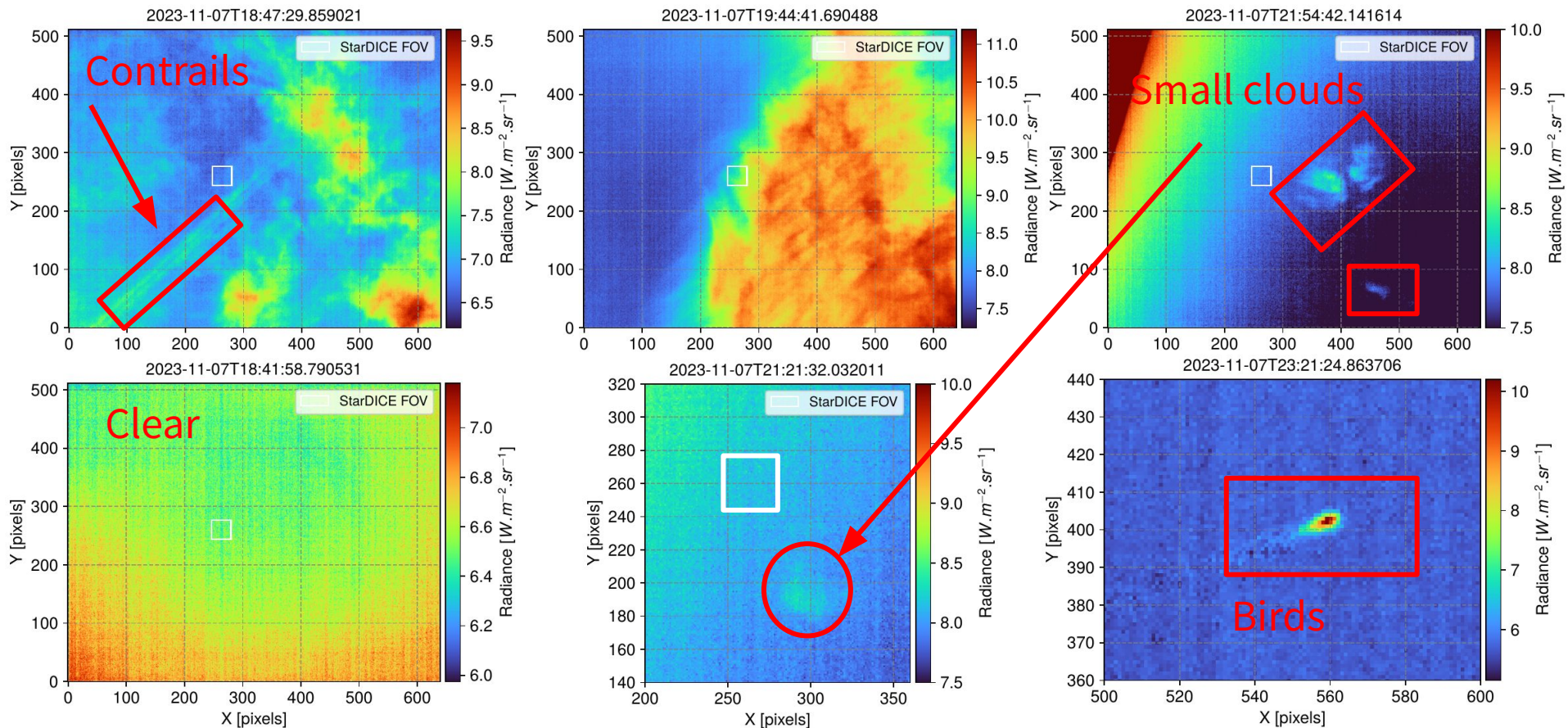


Caveat : extrapolation outside the calibration range (below -30°C)

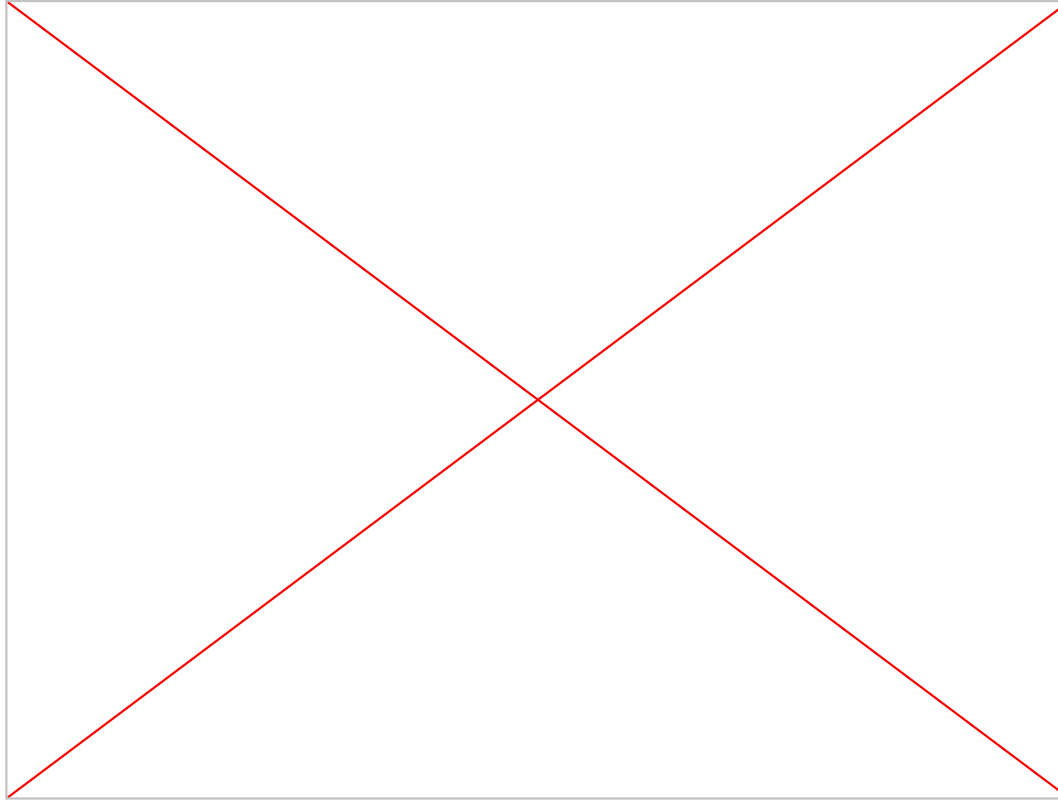


4. Overview of first observations

4.1 Quick look : a lot of things happen in 4h of imaging...



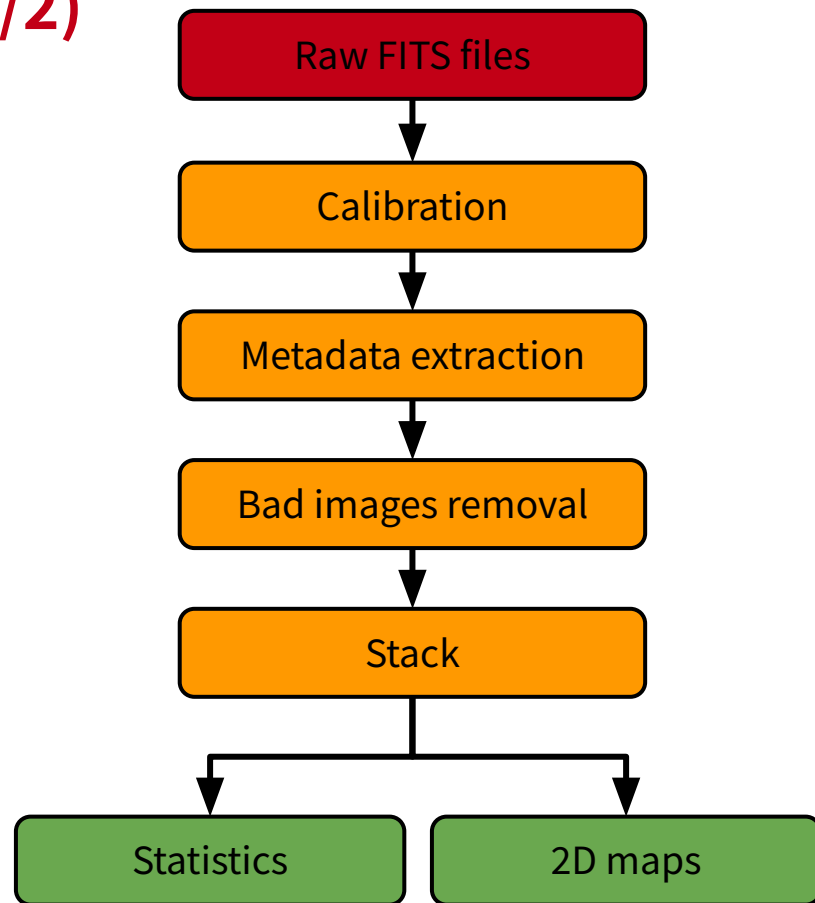
4.2 Night movie animation



<https://drive.google.com/file/d/1ddrEleWxekYS38OtJoGE9FtJotc97m1M/view?usp=sharing>

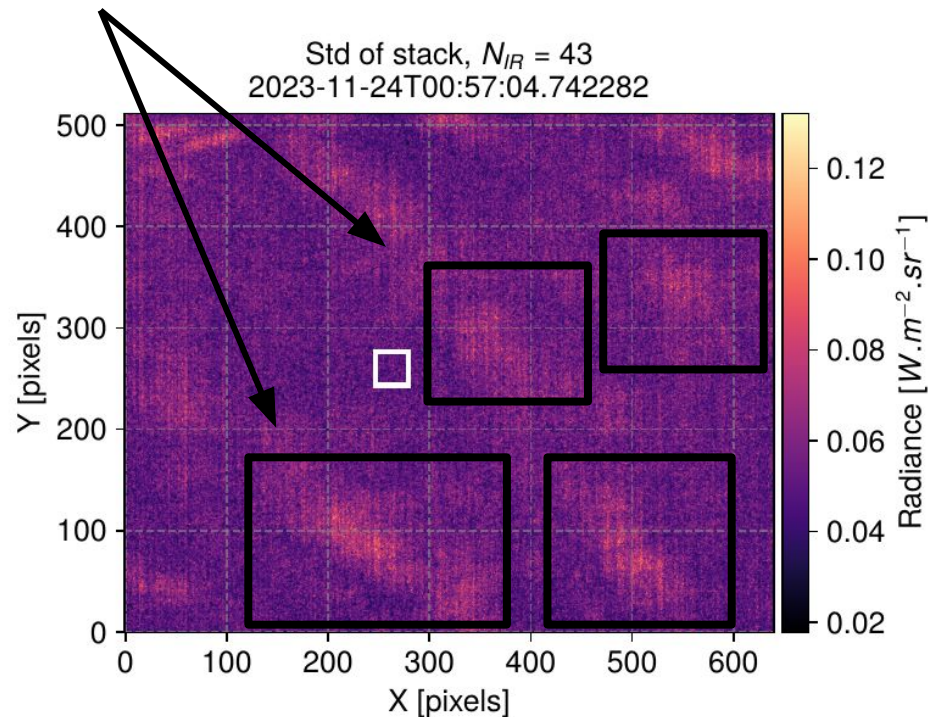
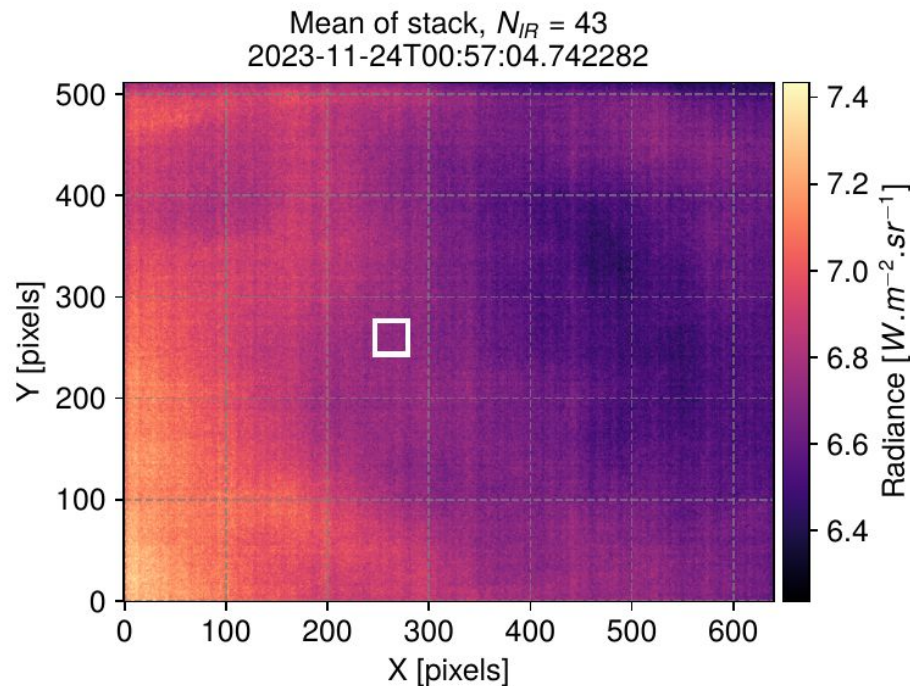
4.3 Pre-processing for DR0 (1/2)

- For each *ugrizG* exposure
⇒ **N_{obs} infrared images**
- Difficult to process all the individual images for analysis
- Statistics : spatial & temporal mean + std of the complete image and the StarDICE FOV
- For now :
 - ~ 80 GB to process per night
 - ~ 1 hours of processing per night on 40-core server



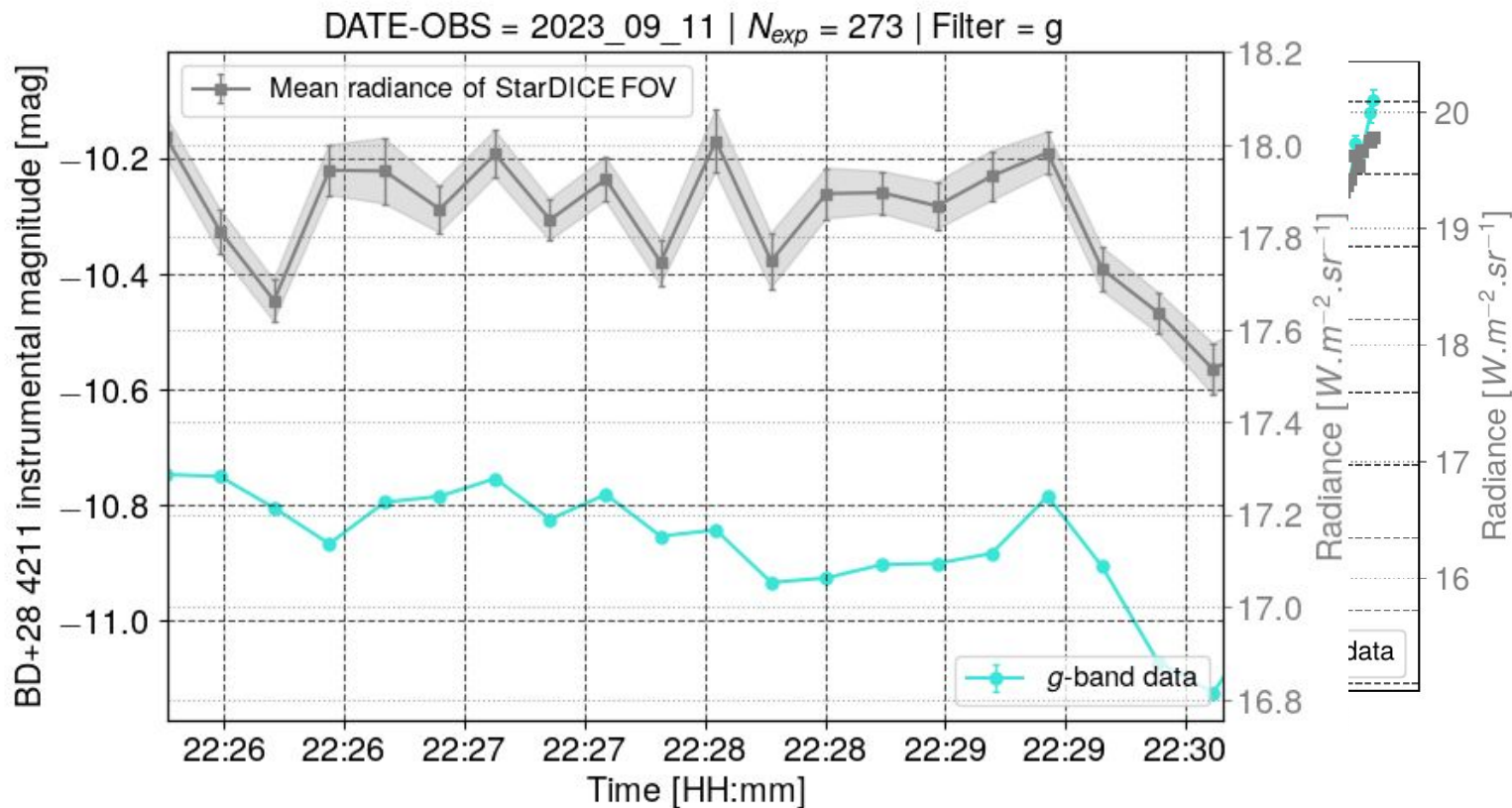
4.3 Pre-processing for DR0 (2/2)

→ Image / cloud structure evolution with time is key



4.4 Preliminary analysis on BD+28 4111 for one night (1/2)

- ~ 1 hour from high variability conditions first test
- Only g-band high-frequency imaging second exposure



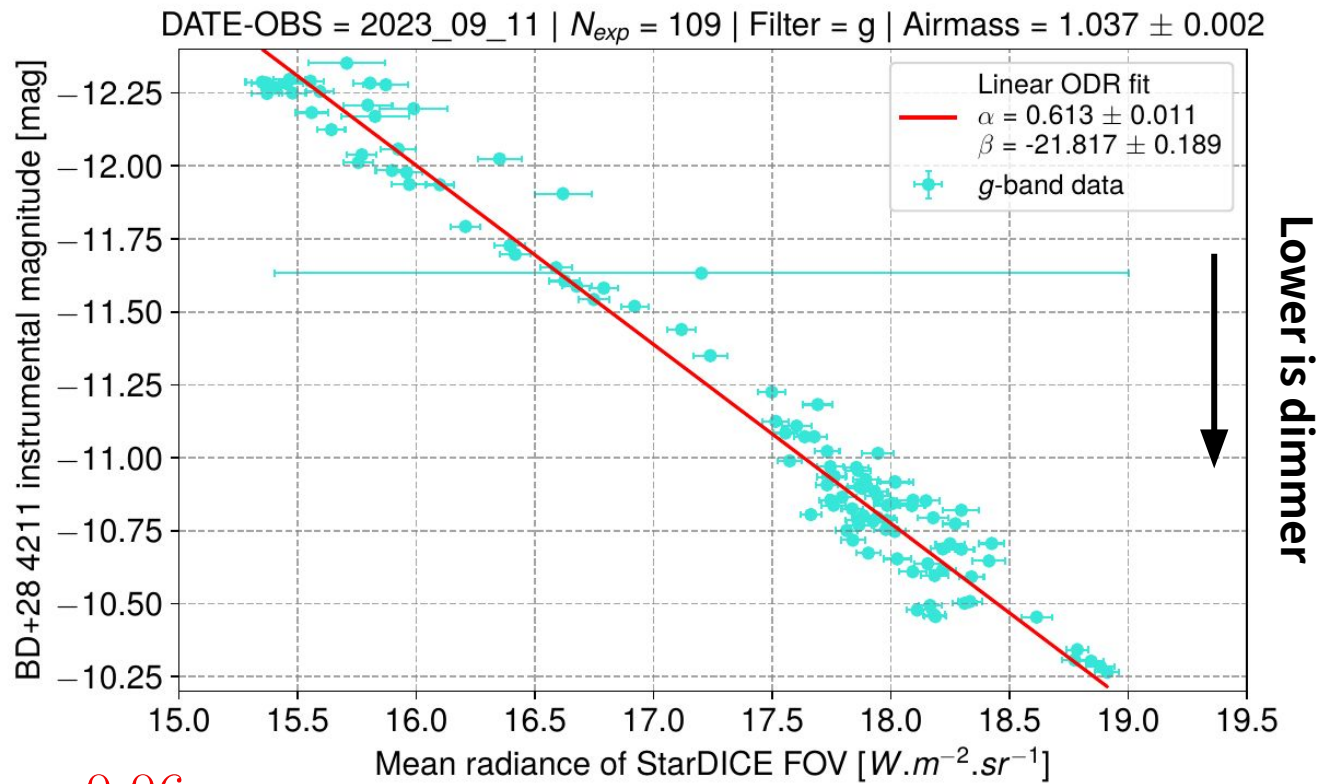
4.4 Preliminary analysis on BD+28 4111 for one night (2/2)

- Strong correlation between magnitude and radiance
- Linear dependence as a first approximation :

$$m_i = \alpha \times L_{i,*} + \beta$$

- With current uncertainty estimation per pixel from calibration :

$$\sigma(m_i) = \alpha \times \sigma(L_{i,*}) \approx 0.06 \text{ mag}$$



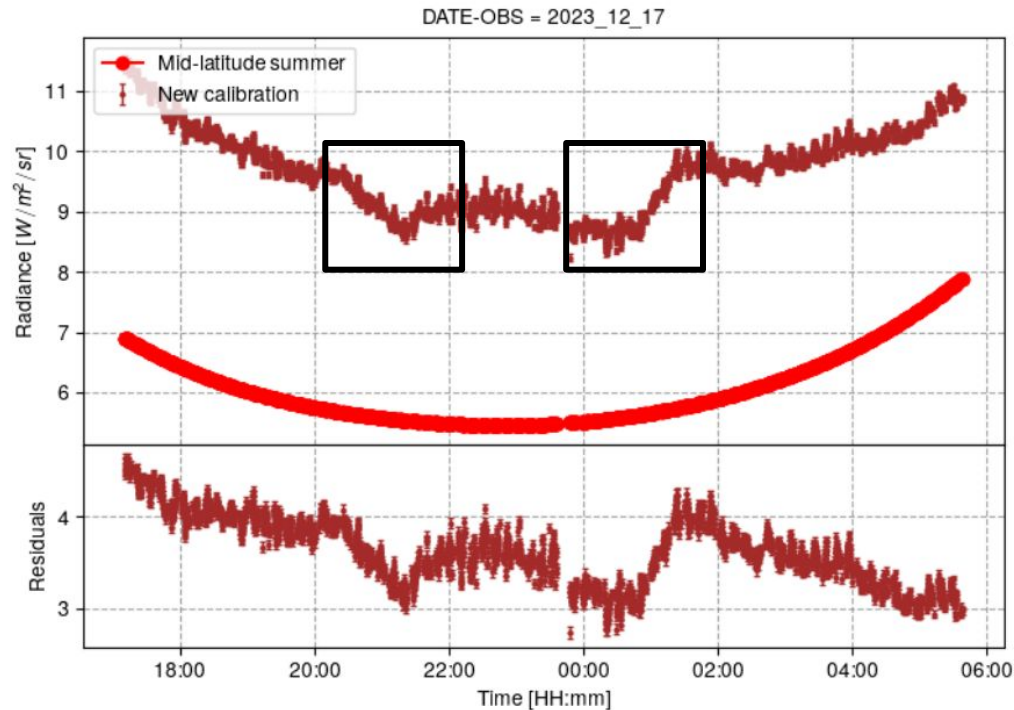
4.5 Caveats

- Dome is thermal emitter itself at $\sim$ ambient temperature
- Variations of dome position relative (known to ~ 1 deg max.) to mount positions

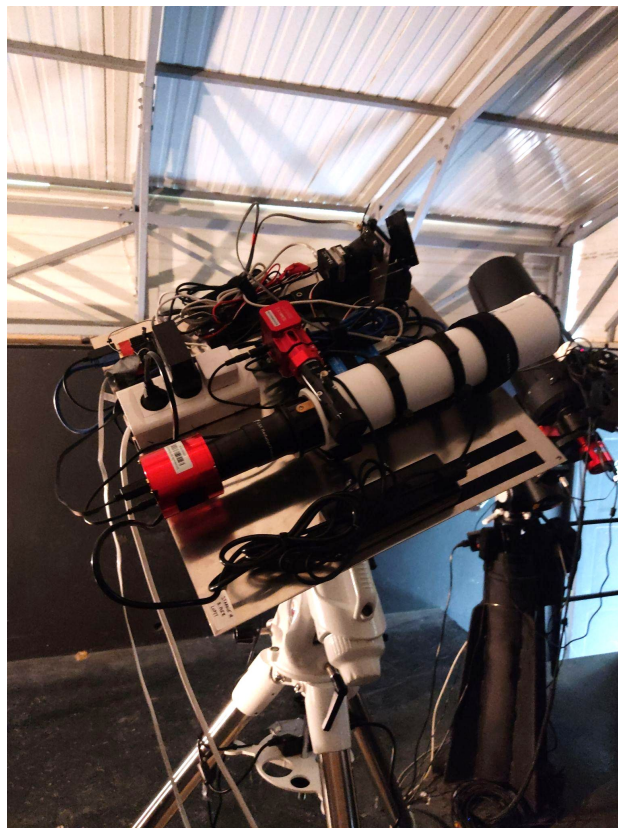
$\Rightarrow$ variations in the background level and structure/gradient in images

$\Rightarrow$ challenge to remove the “True” sky background emission to retrieve cloud(s) structure(s) and optical depth(s)

- **Mitigation plan** : move the IR instrument inside a “roll-off roof” type building



4.6 New setup



Conclusion

- **Summary :**
 - Datasets from StarDICE pre-survey first nights look promising
 - Already +940k images ~ 750 GB
 - ~ 15 nights
 - Remote observations of IR instrument @ OHP
 - Excellent SNR → low spatial and temporal noise on calibrated images
 - Linear dependence between star flux magnitude and IR radiance is confirmed
- **Next steps :**
 - Thorough IR and photometry analysis → robust metric for gray extinction.
 - Combination of IR and UV-VIS-NIR data for atmosphere fitting.
 - Additional data 'without' foreground contamination expected soon
 - Better simulations of the atmosphere with local PWV measurement and atmosphere profiles from radiosondes.
 - Build metric to produce image sky quality map (e.g. combination of time std, spatial std...)